

ENRO EMV 100 mg/ml Oral Solution

Each ml contains:

Active substance: Enrofloxacin: 100 mg

Excipients:

Potassium Hydroxide: 19.0 mg

Propylene glycol: 20.0 mg

Methyl paraben: 1.8 mg

Propyl paraben: 0.2 mg

Purified water q.s to: 1 ml

Product description: clear, yellowish to yellow solution with characteristic odor.

Pharmacodynamics:

Enrofloxacin is a bactericidal agent. The bactericidal activity of enrofloxacin is concentration dependent, with susceptible bacteria cell death occurring within 20–30 minutes of exposure. Enrofloxacin has demonstrated a significant post-antibiotic effect for both gram-negative and -positive bacteria and is active in both stationary and growth phases of bacterial replication. Its mechanism of action is believed to act by inhibiting bacterial DNA-gyrase (a type-II topoisomerase), thereby preventing DNA supercoiling and DNA synthesis. Both enrofloxacin and ciprofloxacin have similar spectrums of activity. These agents have good activity against many gram-negative bacilli and cocci, including most species and strains of *Pseudomonas aeruginosa*, *Klebsiella* spp., *E. coli*, *Enterobacter*, *Campylobacter*, *Shigella*, *Salmonella*, *Aeromonas*, *Haemophilus*, *Proteus*, *Yersinia*, *Serratia*, and *Vibrio* species. Of the currently commercially available quinolones, ciprofloxacin and enrofloxacin have the lowest MIC values for the majority of these pathogens treated. Other organisms that are generally susceptible include *Brucella* spp., *Chlamydia trachomatis*, *Staphylococci* (including penicillinase producing and methicillin-resistant strains), *Mycoplasma*, and *Mycobacterium* spp. (not the etiologic agent for Johne's Disease). The fluoroquinolones have variable activity against most streptococci and are not usually recommended for use in these infections. These drugs have weak activity against most anaerobes and are ineffective in treating anaerobic infections. Bacterial resistance development is an ongoing concern, as many isolates of *Pseudomonas aeruginosa* are now resistant to enrofloxacin. Resistance occurs by mutation, particularly with *Pseudomonas aeruginosa*, *Klebsiella pneumoniae*, *Acinetobacter* and enterococci, but plasmid-mediated resistance is not thought to commonly occur.

Pharmacokinetics:

Enrofloxacin administered via drinking water to poultry is rapidly and very well absorbed with a bioavailability of approx. 90 %. Maximum plasma concentrations of 2 mg/L are reached within 1.5 hours after a single bolus dose rate of 10 mg/kg body weight with a total systemic availability of 14.4 mg.hr/L. Enrofloxacin is eliminated from the body with a total body clearance of 10.3 ml/min.kg. If dosed as continuous drinking water medication (multiple dosing) steady-state concentrations of 0.5 mg (turkeys) to 0.8 mg (chicken) enrofloxacin per liter are achieved. A high mean volume of distribution (5 L/kg) indicated good tissue penetration of enrofloxacin. Concentrations in target tissues like lungs, liver, kidney, intestine and muscle tissue, exceed plasma concentrations by far. In poultry enrofloxacin is poorly metabolised to its active metabolite ciprofloxacin (approximately 5%). Enrofloxacin is eliminated from the body at a half-life of 6 hours. Protein binding in poultry is approximately 25%.

Indication:

Treatment of infections caused by the following bacteria susceptible to enrofloxacin:

Chickens: *Mycoplasma gallisepticum*, *Mycoplasma synoviae*, *Avibacterium paragallinarum*, *Pasteurella multocida*,

Turkeys: *Mycoplasma gallisepticum*, *Mycoplasma synoviae*, *Pasteurella multocida*,

Rabbits

For the treatment infectious diseases due to *Pasteurella multocida* and bacterial enteritis due to infection with *E. coli*.

Recommended dose:

Chickens and Turkeys

10 mg enrofloxacin/kg bodyweight per day for 3-5 consecutive days.

Calculate the daily quantity (ml) of ENRO EMV Oral Solution required for treatment period as follows:

Total number of birds x Average body weight in kg x 0.1= Total volume (ml) per day

ENRO EMV Oral Solution may be put directly into the header tank or introduced via a water proportioner pump.

Rabbits

10 mg/kg bodyweight per day for 5 consecutive days.

Calculate the daily quantity (ml) of ENRO EMV Oral Solution required for treatment period as follows:

Total number of rabbits x Average body weight in kg x 0.1= Total volume (ml) per day

Route of Administration: orally

Contraindication/Warnings/Precautions

Do not use for prophylaxis.

Do not use when resistance/ cross-resistance to (fluoro) quinolones is known to occur in the flock intended for treatment.

Do not use in the case of known hypersensitivity to the active substance, other (fluoro) quinolones or to any of the excipients.

Special warnings and precautions:**Special warnings for each target species and precautions for use**

Treatment of *Mycoplasma spp* infections may not eradicate the organism.

Special precautions for use**Special precautions for use in animals**

Since enrofloxacin was first authorised for use in poultry, there has been widespread reduction in susceptibility of *E. coli* to fluoroquinolones and emergence of resistant organisms. Resistance has also been reported in *Mycoplasma synovia* in the EU.

Special precautions to be taken by the person administering the medicinal product to animals

Those with known hypersensitivity to (fluoro)quinolones should avoid contact with this product.

Avoid contact with skin and eyes.

Rinse any splashes from skin or eyes immediately with water.

Wash hands and exposed skin after use.

Do not eat, drink or smoke while using the product.

Interaction of other medicaments:

The simultaneous application of substances containing aluminium or magnesium can impair the absorption of enrofloxacin.

Pregnancy and lactation:

Do not use in laying hens producing eggs for human consumption.

Do not administer to layer replacement birds within 14 days of coming into lay.

Side effects: None known.

Symptoms and Treatment of overdose:

No adverse clinical symptoms were observed in chickens and turkeys treated respectively with doses up to 10 and 6 times higher than the therapy dose.

Withdrawal Period:

Chickens: Meat and offal: 7 days

Turkeys: Meat and offal: 13 days

Rabbits: Meat and offal: 3 days

Not authorized for use in birds producing eggs for human consumption

Do not administer to layer replacement birds with 14 days of coming in lay.

Storage condition:

The finished product is required to store in the cold place with temperature below 30°C.

Once opened, the medicinal product should be stored at temperatures below 25 °C with closing cap tightly.

Packaging : 100ml and 1L Bottle

Manufacturer:

Emivest Feedmill Vietnam Company Limited-Animal Medicine and Vitamin Plant

Address: No. A-11C-CN Bau Bang Industrial Park, Lai Uyen Commune, Bau Bang District, Binh Duong Province, Vietnam

Product Registration holder, Importer and Distributor:**F.E VENTURE Sdn Bhd**

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